

Calc III: 9.2 Sequences

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- $f(x)^{g(x)} = e^{g(x) \ln f(x)} \Rightarrow \lim_{n \rightarrow a} f(x)^{g(x)} = e^{\left(\lim_{n \rightarrow a} g(x) \ln f(x)\right)}$

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- **Chain rule:** $(f \circ g)'(x) = f'(g(x))g'(x)$

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So then $\lim_{n \rightarrow \infty} \left(\frac{n}{n+5} \right)^n = e^{-5}.$

Squeeze Theorem for sequences

Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ are sequences such that $a_n \leq b_n \leq c_n$ for all $n > N$ for some number N . If $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$ then

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Growth rate of sequences

$\{\ln n\} \ll \{n^p\} \ll \{n^p \ln n\} \ll \{n^{p+s}\} \ll \{b^n\} \ll \{n!\} \ll \{n^n\}$ for $p, s \in \mathbb{R}$ and $b > 1$.

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HW 9.2: 4, 9, 18, 20, 29, 46, 48, 51-56, 58